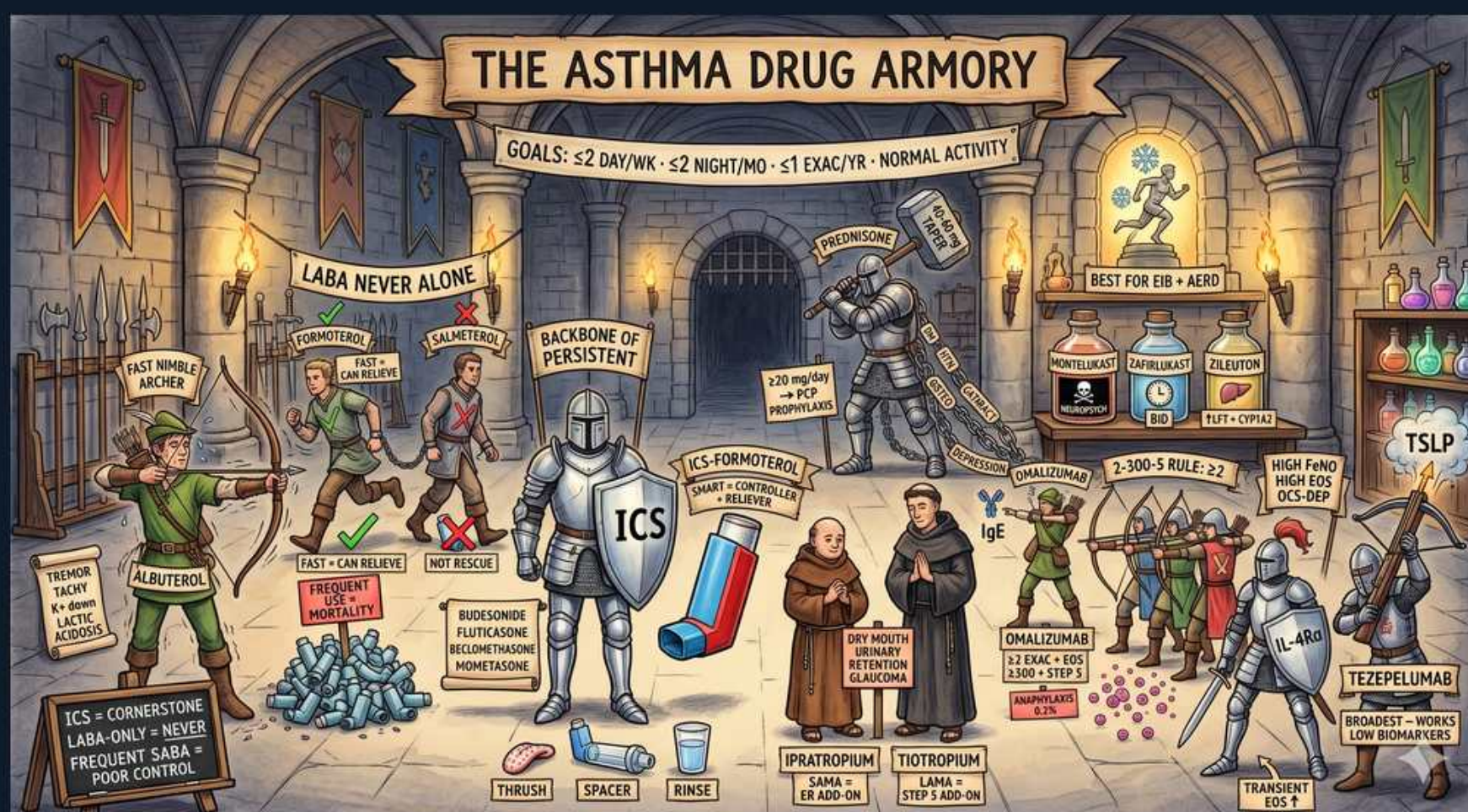


Asthma — The Asthma Drug Armory (drug classes)



1. Control goals banner

WHAT YOU SEE

A banner 'goals: ≤2 day/wk, ≤2 night/mo, ≤1 exac/yr, normal activity'.

WHAT IT ENCODES

Asthma management targets two domains: current SYMPTOM CONTROL (daytime symptoms ≤2/week, no activity limitation, minimal/no nocturnal awakening, reliever need ≤2/week) and future RISK reduction (preventing exacerbations, fixed obstruction, and medication side effects). Control is reassessed at each visit and therapy stepped up or down by the GINA steps.

2. Albuterol archer (SABA reliever)

WHAT YOU SEE

A fast archer 'ALBUTEROL — fast, can relieve' with tremor/tachycardia/lactic acidosis notes.

WHAT IT ENCODES

Albuterol/salbutamol is a short-acting beta-2 agonist (SABA): rapid onset (minutes), used for acute relief and EIB prophylaxis. Side effects from beta stimulation are tremor, tachycardia, palpitations, hypokalemia, and at high doses lactic acidosis and hyperglycemia.

BOARD TRAP

SABA-ONLY treatment is OUT in GINA — overreliance (using >2 canisters/year, or ≥3 inhalers/year markedly raises mortality) signals poor control. The fix is to start/escalate ICS-containing controller therapy, not to keep refilling the reliever.

3. Formoterol (fast LABA, can relieve)

WHAT YOU SEE

A 'FORMOTEROL — fast, can relieve' figure.

WHAT IT ENCODES

Formoterol is a long-acting beta-2 agonist with a UNIQUELY fast onset (minutes, like a SABA) plus long duration (~12 h). This dual property lets ICS-formoterol act as both maintenance and reliever (MART/SMART) — the GINA-preferred reliever across steps 1–5.

4. Salmeterol (LABA, not rescue)

WHAT YOU SEE

A 'SALMETEROL — not rescue' figure.

WHAT IT ENCODES

Salmeterol is a long-acting beta-2 agonist with SLOW onset (~20–30 min), so it is maintenance-only and cannot be used for acute rescue. Always combined with an ICS in a single inhaler.

BOARD TRAP

LABA NEVER ALONE — LABA monotherapy increased asthma deaths (SMART trial), so a LABA must always be paired with ICS; and salmeterol's slow onset means it can never serve as a reliever.

5. ICS — backbone of persistent

WHAT YOU SEE

A knight labeled 'ICS — backbone of persistent' holding an inhaler.

WHAT IT ENCODES

Inhaled corticosteroids (budesonide, fluticasone, beclomethasone, mometasone, ciclesonide) are the controller BACKBONE — they suppress airway inflammation, reduce AHR, prevent exacerbations, and lower mortality. Even mild/intermittent asthma now needs ICS (as-needed low-dose ICS-formoterol).

BOARD TRAP

ICS is the cornerstone of every persistent-asthma regimen; treating with a bronchodilator alone (SABA-only or LABA-only) is never appropriate. Local effects (thrush, dysphonia) are managed with a spacer and mouth rinsing.

6. ICS local effects (thrush/spacer/rinse)

WHAT YOU SEE

Notes 'thrush, spacer, rinse' below the ICS knight.

WHAT IT ENCODES

The common LOCAL adverse effects of ICS are oropharyngeal candidiasis (thrush) and dysphonia/hoarseness. Prevent both by using a spacer and rinsing/gargling and spitting after each dose; switching to a smaller-particle or prodrug ICS (ciclesonide) can also help.

7. ICS-formoterol MART (smart)

WHAT YOU SEE

An 'ICS-FORMOTEROL — smart controller + reliever' inhaler.

WHAT IT ENCODES

ICS-formoterol Maintenance And Reliever Therapy (MART/SMART) uses one inhaler for both scheduled controller dosing and as-needed relief, exploiting formoterol's fast onset. It is GINA's preferred approach — as-needed at Steps 1–2 and as maintenance+reliever at Steps 3–5 — because it reduces severe exacerbations versus SABA-based regimens.

8. Ipratropium / tiotropium (SAMA/LAMA)

WHAT YOU SEE

Two monks 'IPRATROPIUM — ER add-on' and 'TIOTROPIUM — LAMA step 5 add-on'.

WHAT IT ENCODES

Antimuscarinics block M3-mediated bronchoconstriction. Ipratropium (SAMA) is an add-on to SABA in acute SEVERE exacerbations (ED setting). Tiotropium (LAMA, soft-mist) is an add-on controller at Step 4–5 for patients still uncontrolled on ICS-LABA or with a history of exacerbations.

BOARD TRAP

In asthma a LAMA is ADD-ON only — never monotherapy and never a substitute for ICS. (This differs from COPD, where LAMA can be a foundational single agent.)

9. Montelukast / zileuton (LTRA/5-LOX)

WHAT YOU SEE

Bottles 'MONTELUKAST' and 'ZILEUTON — LFT, CYP1A2', best for EIB + AERD.

WHAT IT ENCODES

Leukotriene modifiers reduce cys-LT-driven bronchospasm and mucus. Montelukast/zafirlukast block the CysLT1 receptor; zileuton inhibits 5-lipoxygenase. They are second-line controllers, particularly useful in EIB, AERD, and concomitant allergic rhinitis — but less effective than ICS.

BOARD TRAP

Montelukast carries an FDA neuropsychiatric BOXED warning (mood changes, suicidality); zileuton causes hepatotoxicity (monitor LFTs) and inhibits CYP1A2, RAISING theophylline and warfarin levels.

10. Omalizumab (anti-IgE) archer

WHAT YOU SEE

An archer 'OMALIZUMAB' with '≥2 exac, eos ≥300, step 3, IgE' rule.

WHAT IT ENCODES

Omalizumab (anti-IgE mAb) is a Step 5 add-on biologic for moderate-severe ALLERGIC asthma uncontrolled on ICS-LABA, with documented perennial aeroallergen sensitization and a total IgE within the weight-based dosing range. It lowers free IgE and exacerbation rates.

BOARD TRAP

Omalizumab requires PROVEN allergic sensitization and an IgE level in range — it is the wrong choice for non-allergic eosinophilic asthma (use anti-IL-5) and dosing is set by weight + baseline IgE.

11. Tezepelumab (anti-TSLP)

WHAT YOU SEE

A 'TEZEPELUMAB — broadest, works low biomarkers' shielded figure under 'TSLP'.

WHAT IT ENCODES

Tezepelumab is an anti-TSLP biologic that blocks the upstream epithelial alarmin, suppressing both allergic (Th2) and non-allergic (ILC2) inflammation. As a Step 5 add-on it reduces exacerbations across phenotypes, including those with low eosinophils and low FeNO.

BOARD TRAP

Tezepelumab is the BROADEST biologic — the only one effective regardless of type-2 biomarker level, so it is the answer for severe uncontrolled asthma without high eos/IgE/FeNO.

12. Theophylline / epinephrine (legacy)

WHAT YOU SEE

An 'epinephrine → anaphylaxis' soldier and theophylline references.

WHAT IT ENCODES

Epinephrine (IM) is the treatment of ANAPHYLAXIS — not routine asthma — though it provides nonspecific bronchodilation. Theophylline (a methylxanthine PDE inhibitor/adenosine antagonist) is a legacy controller with a narrow therapeutic window (10–20 µg/mL), CYP1A2 metabolism, and limited modern use.

BOARD TRAP

Do NOT use epinephrine for routine asthma exacerbations (reserve for anaphylaxis); theophylline toxicity causes tachyarrhythmias, seizures, and vomiting and is potentiated by CYP1A2 inhibitors — both are last-resort, not first-line agents.